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CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

INFORMATION TECHNOLOGY

Paper 032 – General Proficiency

ALTERNATIVE TO SBA

1 hour 30 minutes

17 JANUARY 2012 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of FOUR questions.
2. Answer ALL questions.
3. Number EACH answer correctly in the answer booklet.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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Answer ALL questions.

1. WORD PROCESSING

Consider the letter shown below.

NT WIRELESS

Middle Street, Georgetown, Guyana
Tel: 229-8172; Email: info@ntwireless.com

December 20, 2011

Ms Allan Monroe
10 High Street
Georgetown

Dear Ms Monroe:

Thank you for your patronage in 2011. I am pleased to inform you that we have completed our network upgrade with state-of-the-art technologies. Effective 1 January, 2012, we will be offering the following services at drastically reduced rates:

- DSL for both residential and commercial customers
- WI-FI Internet hotspots at many locations
- WIMAX Internet access

Please check our website, www.ntwireless.com, or contact our sales department for more information on our services and rates. We are currently accepting applications for our new services.

Have a prosperous New Year and enjoy our new services.

Yours sincerely,

Carol Smith
Marketing Director

Bringing State-of-the-Art Technologies to the Nation

GO ON TO THE NEXT PAGE

- (a) What justification has been used on the FIRST paragraph of the body of the letter? **(1 mark)**
- (b) Which of the following is used in the letter?
- Header
 - Footnote
 - Footer
- (1 mark)**
- (c) What line spacing has been used on the SECOND paragraph of the body of the letter? **(1 mark)**
- (d) How can the hyperlink (underline) in the website address be removed? **(2 marks)**
- (e) For emphasis, the website address must be put in RED. Explain how this is done. **(2 marks)**
- (f) Write THREE formatting features used on the name of the company (NT WIRELESS). **(3 marks)**
- (g) The secretary needs to insert the company's logo located on a flash drive into the letter to the left of the name and address of the company. List the steps required to do this. **(3 marks)**
- (h) State what EACH of the following icons is used for in a word processing application:
- (i)  (i) 
- (ii)  (ii) 
- (iii)  (iii) 
- (3 marks)**
- (i) The secretary wishes to send this letter to other customers with their names and addresses included in the letter using mail merge. Name TWO documents which are required to perform a mail merge. **(2 marks)**
- (j) State, with a reason, the number of merge fields that should be in the letter. **(2 marks)**

Total 20 marks

2. SPREADSHEET

The following spreadsheet shows the population of elderly persons in a country for the year 2011. Use it to answer the questions that follow.

	A	B	C	D	E	F
1	Region	60 - 69	70 - 79	80 & over	Total	% of Total
2	1	7,600	6,000	4,900	18,500	21
3	2	9,500	9,800	8,500	27,800	32
4	3	7,500	4,800	5,600	17,900	21
5	4	4,100	3,900	2,600	10,600	12
6	5	2,800	3,800	5,400	12,000	14
7	TOTAL	31,500	28,300	27,000	86,800	

- Give the range of cells that contain all elderly persons 70 years and over. **(1 mark)**
- What formatting feature has been applied to all numeric data? **(1 mark)**
- What alignment has been applied to the headings of all numeric columns? **(1 mark)**
- Write functions or formulae to calculate EACH of the following:
 - The total number of elderly persons in the country who are 80 years and over **(2 marks)**
 - The minimum number of elderly persons, 70–79 years old, in a region **(2 marks)**
 - The average of the regional totals of elderly persons **(2 marks)**
- If the data in the spreadsheet is sorted on the total for each region in descending order, which region would be at the TOP of the sorted list? **(1 mark)**
- State the spreadsheet feature that would cause Row 1 to remain visible on the spreadsheet as the users scroll down the rows. **(1 mark)**
- If the formula, =B7/\$M\$2 is entered in a cell and the formula is copied to another cell, what will happen to the values in the numerator and the denominator? **(2 marks)**
- Complete the criteria range below to filter the data for ALL regions with over 5,000 elderly persons in EACH age group.

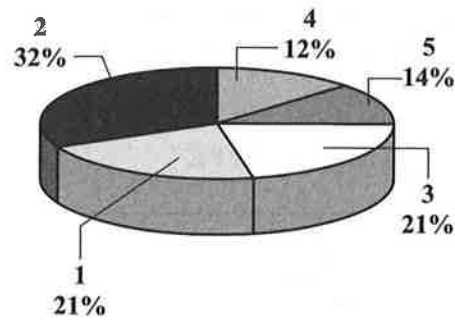
Region	60–69	70–79	80 & over

(2 marks)

- State the region that will be extracted from Part (h). **(1 mark)**

GO ON TO THE NEXT PAGE

- (j) The following chart was prepared from the data in the spreadsheet.



- (i) What is the name of the chart? (1 mark)
- (ii) State the ranges that were used to create the chart. (2 marks)
- (iii) What is the purpose of the chart? (1 mark)

Total 20 marks

3. DATABASE MANAGEMENT

The Ministry of Home Affairs of a country maintains a database with two tables to store information on the crimes committed in 2011. The two tables and a report generated from the tables are listed below.

TABLE 1: TYPE

CRIME_ID	TYPE
F20	Fraud
V35	Vehicle Theft
R25	Robbery
B30	Burglary
M30	Murder

TABLE 2: DIVISION

CRIME_ID	DIVISION	REPORTED CASES	CONVICTION
B30	A	60	40
B30	B	120	100
F20	A	25	15
F20	??	??	??
M30	C	7	5
M30	B	12	7
V35	C	5	3
V35	A	15	10
V35	B	25	20
R25	C	25	20
R25	??	??	??
R25	B	175	140

- (a) State the data type for the CRIME_ID and the CONVICTION fields in the DIVISION TABLE. **(2 marks)**
- (b) What is an appropriate **primary** key for the TYPE table? **(1 mark)**

Part 3 (c) and 3 (d) are based on the following report.

CRIME STATISTICS 2011			
TYPE	Reported cases	Division	Conviction
<i>Burglary</i>			
	60	A	40
	120	B	100
Summary for 'TYPE' = Burglary (2 detail records)			
Sum	180		140
<i>Fraud</i>			
	25	A	15
	75	B	50
Summary for 'TYPE' = Fraud (2 detail records)			
Sum	100		65
<i>Murder</i>			
	7	C	5
	12	B	7
Summary for 'TYPE' = Murder (2 detail records)			
Sum	19		12
<i>Robbery</i>			
	25	C	20
	83	A	78
	175	B	140
Summary for 'TYPE' = Robbery (3 detail records)			
Sum	283		238
<i>Vehicle Theft</i>			
	5	C	3
	15	A	10
	25	B	20
Summary for 'TYPE' = Vehicle Theft (3 detail records)			
Sum	45		33
Grand Total	627		488
Tuesday, 03 January, 2012		Page 1 of 1	

(c) Complete the missing data in the sections shaded in the DIVISION table (Table 2 on page 6). **(2marks)**

(d) State:

(i) The title of the report **(1 mark)**

(ii) The field on which the records were grouped **(1 mark)**

(iii) The fields on which the report was sorted **(2 marks)**

(iv) The total number of reported cases of crimes and the total number of convictions **(2 marks)**

(v) The field used to join Tables 1 and 2. **(1 mark)**

GO ON TO THE NEXT PAGE

- (e) A calculated field named "ROC" is to be created in the DIVISION table to store the percentages of reported cases that resulted in convictions. State the formula for the calculated field. **(2 marks)**
- (f) You wish to write a query to show all divisions with more than 50 convictions.
- (i) Write the appropriate field names needed for the query. **(2 marks)**
 - (ii) Write the query. **(2 marks)**
 - (iii) Write the result of the query. **(2 marks)**

Total 20 marks

4. PROGRAMMING AND PROBLEM SOLVING

- (a) Consider the fragment of Pascal code below.

Line
Number

```
1      Tax_Rate = 0.10;
2      repeat
3          read (Cost_of_Game);
4          MISC := Cost_of_Game * Tax_Rate;
5          Price = Cost_of_Game + MISC;
6          Writeln ('Please pay': ',Cost_of_Game');
7      until Cost_of_Game = -99;
```

- (i) Identify the constant. **(1 mark)**
- (ii) Write an appropriate data type for Cost_of_Game. **(1 mark)**
- (iii) Write an appropriate variable name to replace MISC. **(1 mark)**
- (iv) After the program is executed, explain where the statement in Line 6 will cause the cursor to be placed. **(1 mark)**
- (v) Explain the purpose of using -99 in Line 7. **(2 marks)**
- (vi) Draw a flowchart to represent the fragment of code given. **(8 marks)**
- (vii) For EACH of the following, write 'Yes' AND the line number(s) if the following is found in the code above, otherwise write 'No':

- P1: A logic operation
- P2: An arithmetic operation
- P3: A conditional statement
- P4: A loop

(6 marks)

GO ON TO THE NEXT PAGE

- (b) (i) Copy and complete the following truth table which determines the conditions for completing a subject. Note that 1 represents 'To Pass' and 0 represents 'To Fail'.

SBA	Exam	SBA AND Exam
1	1	
1	0	
0	1	
0	0	

(4 marks)

- (ii) Write the TWO conditions that are necessary for passing a subject. (1 mark)

- (c) Answer the question below based on the following array called LIST.

LIST	C	A	R	I
INDEX →	1	2	3	4

Write a fragment of code to search for the letter 'R' in the array, and output the index where the letter is found. (Assume that the array has been declared). (5 marks)

Total 30 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.